

REDEMPTION CITY NAVIGATOR

Architecture & System Design Document

Offline-first navigation, location discovery and local service access for Redemption City.

Executive Summary

Redemption City Navigator is a working mobile MVP built to make movement inside Redemption City clearer, faster and safer. The app brings together offline-first place discovery, internal route guidance, chalets access, Market Hub, Sunday Shuttle and service discovery in one mobile experience. It is designed for residents, visitors, worshippers, vendors and first-time camp attendees who need reliable direction without depending fully on asking strangers or searching scattered information.

Positioning Statement

A practical navigation and local-services layer for Redemption City - built around verified places, saved road corridors, mobile-first usability and offline resilience.

PRODUCT	Redemption City Navigator
SUBMISSION CATEGORY	Safety / Navigation / Local Services
CORE PLATFORM	Expo React Native mobile app with TypeScript
CURRENT STATE	Working mobile MVP with searchable local data, route display foundation, tab navigation and service modules.
PRIMARY USERS	Visitors, worshippers, residents, vendors, chalet users, shuttle users and camp operators.
REPOSITORY	https://github.com/Supreme5050/redemption-city-navigator

1. Problem and Product Value

Redemption City is large, active and highly dynamic. Many people arrive for services, events, conventions, hostel stays, shopping, meetings and family visits, yet they often struggle to identify the right roads, chalets, vendors and landmarks. The result is wasted time, repeated questions, missed appointments, poor visitor experience and avoidable stress.

The Navigator solves this by turning local knowledge into a structured mobile system. Instead of relying on memory, assumptions or verbal directions, users can search for a destination, view mapped routes, discover vendors, access chalet information and use mobility features from a single app interface.

Why It Scores Well

The solution connects a real campus-scale navigation pain point with a usable mobile prototype, local datasets, offline-first thinking and a path toward operational adoption inside Redemption City.

2. Product Readiness Snapshot

- Mobile app runs in Expo and is structured as a working React Native prototype.
- Core navigation screens are implemented with Expo Router tab navigation.
- Local place search is connected to structured Redemption City place data.
- Map screen displays satellite/hybrid 3D view with route lines, destination markers and current-location marker.
- Verified route data is separated into a dedicated route file for clean improvement and maintenance.
- Market, chalet, shuttle, vendor and profile modules exist as product sections for review.
- GitHub repository is public and contains the source code for technical review.
- The architecture supports quick updates as additional road and junction coordinates are verified.

3. Core User Experience

1. A user opens the mobile app and lands on the main navigation experience.
2. The user searches for a road, building, church, chalet, vendor or local service.
3. The app resolves the search from its local place dataset.
4. The map displays the destination marker and the saved road corridor for that destination.
5. The user can switch into satellite/hybrid 3D view for better physical context.
6. Additional tabs give access to Market Hub, Chalets, Sunday Shuttle and profile/service areas.

4. System Architecture

The system is organized around a clean separation of presentation, local data, map rendering and service modules. This makes the app reviewable, maintainable and easy to extend without breaking existing screens.

Layer	Responsibility	Implementation
Presentation	Mobile screens, tab navigation, map view, marketplace, chalet and shuttle flows.	Expo Router, React Native screens, typed components, reusable UI elements.
Location & Map	Searches places, displays current position, renders internal route corridors in hybrid/3D map view.	react-native-maps, Expo Location, Google map provider, local route coordinates.
Offline Data	Keeps important Redemption City places available without depending on live internet queries.	Static TypeScript datasets: places, routes, vendors, chalets, products.
Routing Logic	Selects the best saved corridor for a destination and draws a route polyline from verified road points.	verifiedRoutes.ts plus route selection and coordinate-calibration workflow.
Service Modules	Supports visitor navigation, market discovery, chalet lookup, Sunday Shuttle and vendor activity.	Modular screens connected to typed local datasets and ready for API/database sync.
Deployment Evidence	Makes the project reviewable through source code, mobile demo, screenshots and PDFs.	GitHub repository, Expo prototype, Drive links, submission documents.

High-Level Data Flow

User search -> places.ts -> selected destination -> verifiedRoutes.ts -> route coordinates -> MapView Polyline + markers -> mobile navigation experience

Architecture Advantage

The app does not depend on a remote map API for local place discovery. Redemption City knowledge is structured inside the project and can be improved with verified coordinates over time.

5. Capability Modules

Capability	What It Delivers	Data / Logic Layer	Review Evidence
Map & Navigation	Shows 3D/hybrid map, destination marker, current location and saved internal route corridor.	places.ts, verifiedRoutes.ts, map screen logic.	Screenshots show route selection, satellite view and markers.
Offline Place Search	Searches roads, landmarks, churches, offices and key service points.	Typed local place dataset with aliases and address hints.	Search bar resolves Heritage Road and other locations.
Chalets	Gives users a dedicated area for accommodation discovery and future hospitality	Chalet screen and dataset structure.	Tab visible in mobile prototype.

Capability	What It Delivers	Data / Logic Layer	Review Evidence
	integration.		
Market Hub	Organizes local commerce discovery and vendor/customer interaction direction.	Vendor, products and market screens.	Market tab and vendor screens exist in source code.
Sunday Shuttle	Supports Sunday movement pain point by separating mobility into a focused feature area.	Shuttle screen and route-ready product structure.	Source includes Sunday Shuttle flow.
Vendor Center	Provides structure for sellers, orders, products and vendor operations.	Vendor dashboard, wallet, orders and product screens.	Multiple vendor files present in app source.
Feedback & Proof Flows	Supports user trust through feedback, delivery proof and protected order screens.	Modular screens for order proof, dispute and feedback.	Files exist as part of the mobile app structure.

6. Offline-First Design

The MVP treats local knowledge as a core asset. Important places, route corridors, vendors and service categories are kept inside the app as structured TypeScript data. This means the app can provide discovery and route context even when internet quality is weak or unreliable.

- Local datasets reduce dependence on live search results.
- Aliases help users find locations even when they type shortened names or local variants.
- Route coordinates are stored separately from UI code, making accuracy improvements safe and controlled.
- The map layer remains flexible: standard, satellite and hybrid views can be used depending on the demonstration need.
- The app is ready for database synchronization while retaining its offline-first foundation.

Coordinate Accuracy Strategy

The route system is built around verified road and junction points. Coordinates are managed as route corridors, so new road bends can be added without redesigning the map interface.

7. Navigation and Route Model

The map experience uses a saved-corridor approach. A destination is selected from the local place dataset, the best matching route is selected from verifiedRoutes.ts, and the route is drawn as a polyline over the map. This approach is practical for a private or semi-private environment where public map data may be incomplete.

ROUTE SOURCE	verifiedRoutes.ts stores named road corridors and their coordinate sequences.
DESTINATION SOURCE	places.ts stores places, aliases, zones and location metadata.
MAP RENDERING	react-native-maps displays route polylines, user position, saved route start and destination marker.
ACCURACY WORKFLOW	Road bends and junctions are calibrated directly from satellite/hybrid map view and added as route points.
NAVIGATION LOGIC	The route selector prioritizes the most relevant saved corridor for each destination.

8. Review Readiness and Demonstration Evidence

The project is ready for review as a mobile MVP. The submission package includes source code, architecture documentation, business documentation, screenshots and a prototype demonstration link. This gives reviewers both technical evidence and product evidence.

- Public GitHub repository confirms code ownership and implementation work.
- Mobile demo video shows the app running on a phone through Expo.
- Screenshots demonstrate the actual interface, map, tabs and service modules.
- This architecture document explains the system, modules and data flow.
- Business/GTM document explains impact, adoption and sustainability.
- Pitch deck summarizes the opportunity in a format judges can scan quickly.

Submission Message

Redemption City Navigator is not just a concept. It is a working mobile prototype backed by a structured codebase, offline-ready local data and a scalable architecture for a real Redemption City navigation and services ecosystem.

9. Technical Stack

MOBILE FRAMEWORK	Expo React Native
LANGUAGE	TypeScript
NAVIGATION	Expo Router
MAP LAYER	react-native-maps with Google provider
LOCATION	expo-location
DATA MODEL	Typed local datasets for places, routes, vendors, products and chalets
VERSION CONTROL	Git and GitHub
PROTOTYPE DELIVERY	Expo mobile demo, screenshots and repository review

10. Impact, Scalability and Why This Matters

The value of the product is strongest because it solves a daily, visible and repeatable problem in a real location. Redemption City receives people who need direction, accommodation, food, shopping, church locations, transport support and service access. A dedicated navigation platform can reduce confusion while strengthening local commerce and visitor confidence.

- Visitors reach destinations faster and with less confusion.
- Residents and worshippers gain a structured local discovery tool.
- Vendors become more discoverable through the Market Hub direction.
- Chalet access becomes easier to organize and communicate.
- Sunday movement can be coordinated through a dedicated shuttle layer.
- The same architecture can support other large religious camps, estates, universities and event grounds.

Closing Statement

Redemption City Navigator is a strong MVP because it combines a real pain point, a working mobile product, a clear technical structure and a scalable local-services model. It is built to move from hackathon prototype into practical deployment.

11. Submission Links Placeholder

Use this section to keep the final links organized before pasting them into the submission portal.

GITHUB REPOSITORY	https://github.com/Supreme5050/redemption-city-navigator
WORKING PROTOTYPE / DEMO VIDEO	Google Drive public link to be inserted
SCREENSHOTS FOLDER	Google Drive public link to be inserted
ARCHITECTURE PDF	Google Drive public link to be inserted
BUSINESS / GTM PDF	Google Drive public link to be inserted
PITCH DECK PDF	Google Drive public link to be inserted